MTH301A: *Analysis I*
Homework: These will be periodically updated. Odd Semester, 2023/24

(1) Read the proofs of Propositions 1.14, 1.15, 1.16 and 1.18 in Rudin's Chapter 1.

(2) Let $(F, \oplus, \cdot, \prec)$ be an ordered field. Show that F contains an isomorphic copy of the ordered field of rationals. More precisely, show that there is an injection $h : \mathbb{Q} \to F$ such that the following hold for every $x, y \in \mathbb{Q}$.

   (a) $h(x + y) = h(x) \oplus h(y)$

   (b) $h(xy) = h(x) \cdot h(y)$

   (c) $h(0) = 0_F$ and $h(1) = 1_F$

   (d) $x < y \iff h(x) \prec h(y)$

(3) Let $\prec$ be any linear order on $\mathbb{C}$ (the field of complex numbers). Show that $(\mathbb{C}, +, \cdot, \prec)$ is not an ordered field.

(4) Read the Appendix at the end of Chapter 1 in Rudin's book.

(5) Show that any two complete ordered fields are isomorphic. This means that if $(F_1, +_1, \cdot_1, <_1)$ and $(F_2, +_2, \cdot_2, <_2)$ are both complete ordered fields, then there is a **bijection** $h : F_1 \to F_2$ such that the following hold for any $x, y \in F_1$.

   (a) $h(x +_1 y) = h(x) +_2 h(y)$

   (b) $h(x \cdot_1 y) = h(x) \cdot_2 h(y)$

   (c) $h(0_{F_1}) = 0_{F_2}$ and $h(1_{F_1}) = 1_{F_2}$

   (d) $x <_1 y \iff h(x) <_2 h(y)$

(6) Show that a set A is countable iff there is an injective function $f : A \to \mathbb{N}$.

(7) A real number $x \in [0, 1]$ is said to be computable if there is a computer program that on all inputs $n \geq 1$ halts and outputs the nth decimal digit of x. Show that the set of computable real numbers is countable.

(8) An algebraic real number is a real number which is the root of a polynomial equation

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$$

with integer coefficients ($\{a_0, a_1, \cdots, a_n\} \subseteq \mathbb{Z}$ and $a_n \neq 0$). Show that the set of all algebraic real numbers is countable.

(9) Let $f : \mathbb{R} \to \mathbb{R}$ be a increasing function. Show that $\{a \in \mathbb{R} : f \text{ is discontinuous at } a\}$ is countable.

(10) Let $C(\mathbb{R})$ be the set of all continuous functions from $\mathbb{R}$ to $\mathbb{R}$. Show that $|C(\mathbb{R})| = |\mathbb{R}|$. Hint: First show that if $f, g \in C(\mathbb{R})$ and $f \restriction \mathbb{Q} = g \restriction \mathbb{Q}$, then $f = g$. Then use the fact that the set of all functions $h : \mathbb{Q} \to \mathbb{R}$ has cardinality equal to $|\mathbb{R}^{\mathbb{N}}| = |\mathbb{R}|$.

(11) For every set X, show that $|X| < |\mathcal{P}(X)|$. This means that there is an injection from X to $\mathcal{P}(X)$ but there is no injection from $\mathcal{P}(X)$ to X.

(12) Prove the Cauchy-Schwarz inequality in $\mathbb{R}^n$. For all $\vec{x}, \vec{y} \in \mathbb{R}^n$,

$$|\vec{x} \cdot \vec{y}| \leq ||\vec{x}||||\vec{y}||.$$

Recall that $\vec{x} \cdot \vec{y} = \sum_{1 \leq i \leq n} x_i y_i$ and $||\vec{x}|| = (\vec{x} \cdot \vec{x})^{1/2}$ where $\vec{x} = (x_1, x_2, \cdots, x_n)$ and $\vec{y} = (y_1, y_2, \cdots, y_n)$.

(13) Show that $d(\vec{x}, \vec{y}) = ||\vec{x} - \vec{y}||$ defines a metric on $\mathbb{R}^n$.

(14) Consider $\mathbb{R}$ under the usual metric ($d(x, y) = |x - y|$). Let $E \subseteq \mathbb{R}$ be as in (a)-(e) below. Compute E' (the set of limit points of E), $\mathsf{cl}(E)$ (the closure of E) and $\mathsf{Int}(E)$ (the interior of E) in each case.

(a) $E = (0, 1)$.

(b) $E = [0, 1)$.

(c) $E = \mathbb{Z}$.

(d) $E = \mathbb{Q}$.

(e) $E = \{1/n : n \geq 1\}$.

(15) Construct a bounded subset $E \subseteq \mathbb{R}$ such that $|E'| = 5$ (i.e., E has exactly 5 limits points in $\mathbb{R}$).

(16) Let (X, d) be a metric space and $E \subseteq X$. Show that $E'' \subseteq E'$. Conclude that E' is closed in X.

(17) Try problems 1-11 (as many as you can) in Rudin Chapter 2.

(18) Let (X, d) be a metric space and $E \subseteq X$. Show that the following are equivalent.

(a) E is closed in X.

(b) For every sequence $\langle x_n : n \geq 1 \rangle$ of points in E, if $\langle x_n : n \geq 1 \rangle$ converges to $y \in X$, then $y \in E$.

(19) Let (X, d) be a metric space and $E \subseteq X$. Show that the following are equivalent.

(a) (X, d) is a complete metric space.

(b) For every sequence $\langle C_n : n \geq 1 \rangle$ of closed balls in X, if $C_{n+1} \subseteq C_n$ for all $n \geq 1$ and if radius of C_n goes to zero as $n \to \infty$, then $\bigcap_{n \geq 1} C_n \neq \emptyset$.

(20) Show that $\mathbb{R}$ is a complete metric space under the usual metric. Hint: You can use the previous problem or argue directly as follows. Let $\langle x_n : n \geq 1\rangle$ be a Cauchy sequence in $\mathbb{R}$. Define $a_n = \inf(\{x_k : k \geq n\})$ and $b_n = \sup(\{x_k : k \geq n\})$. Check that $a_n \leq a_{n+1} \leq b_{n+1} \leq b_n$ for every $n \geq 1$. Next show that $\lim_{n\to\infty}(b_n - a_n) = 0$. Define $x = \sup(\{a_n : n \geq 1\})$. Show that $\lim_{n\to\infty} x_n = x$.

(21) Show that $\mathbb{R}^n$ is complete metric space under the usual metric. Hint: Use the fact that $\mathbb{R}$ is a complete metric space.

(22) Show that the conclusion of the Baire category theorem doesn't hold if we do not assume that (X, d) is complete. Hint: Show that it fails for $X = \mathbb{Q}$.

(23) Let $\{a_n : n \geq 1\}$ be a one-to-one list of the set all rationals $\mathbb{Q}$. For each $n \geq 1$, define

$$U_n = \bigcup_{k\geq 1} \left(a_k - 2^{-(n+k)}, a_k + 2^{-(n+k)}\right).$$

Is $\bigcap_{n\geq 1} U_n = \mathbb{Q}$? Justify your answer.

(24) Show that the intersection of any family of compact subsets of a metric space (X, d) is compact.

(25) Suppose (X, d) is a metric space and $\mathcal{F}$ is a family of compact subsets of X. Assume that for every finite $\mathcal{E} \subseteq \mathcal{F}$, $\bigcap \mathcal{E} \neq \emptyset$. Show that $\bigcap \mathcal{F} \neq \emptyset$.

(26) Show that every compact metric space is both complete and bounded.

(27) Give an example of a complete and bounded metric space which is not compact.

(28) Let $a_k < b_k$ for $k \in \{1, 2, \cdots, n\}$. Show that $[a_1, b_1] \times \cdots \times [a_n, b_n]$ is compact under the Euclidean metric.

(29) Suppose (X, d) is a compact metric space and $f : X \to \mathbb{R}$ is continuous. Show that there are $x_0, x_1 \in X$ such that f attains its maximum value at x_0 and f attains its minimum value at x_1.

(30) Suppose (X, d) is a connected metric space and $f : X \to \mathbb{R}$ is continuous. Assume that $x, y \in X$ and $f(x) < f(y)$. Show that for every $a \in [f(x), f(y)]$, there exists $z \in X$ such that $f(z) = a$.

(31) Suppose (X, d_1) and (Y, d_2) are metric spaces and $f : X \to Y$. Show that the following are equivalent.

(a) f is uniformly continuous on X.

(b) For every sequence $\langle (x_n, y_n) : n \geq 1\rangle$ of pairs of points in X,

$$\lim_{n\to\infty} d_1(x_n, y_n) = 0 \implies \lim_{n\to\infty} d_2(f(x_n), f(y_n)) = 0.$$

(32) Let $f : \mathbb{R} \to \mathbb{R}$. Define

$$\mathsf{osc}(f, x) = \lim_{h \to 0} \sup(\{|f(a) - f(b)| : a, b \in (x - h, x + h)\})$$

(a) Show that f is continuous at x iff $\mathsf{osc}(f, x) = 0$.

(b) Show that for each $a > 0$, the set $\{x \in \mathbb{R} : \mathsf{osc}(f, x) < a\}$ is open in $\mathbb{R}$.

(c) A subset $E \subseteq \mathbb{R}$ is a G_δ (G-delta) set if it is the intersection of some countable family of open sets. Show that $\{x \in \mathbb{R} : f \text{ is continuous at } x\}$ is a G_δ set.

(d) Show that $\mathbb{Q}$ is not a G_δ set.

(e) Show that there is no function $f : \mathbb{R} \to \mathbb{R}$ that is continuous at every rational and discontinuous at every irrational.

(33) Let C denote the Cantor set. Show that $C' = C$. Also show that C is nowhere dense in $\mathbb{R}$.

(34) We say that $E \subseteq \mathbb{R}$ has measure zero iff for every $\varepsilon > 0$, there exists a sequence of open intervals $\langle J_n : n \geq 1 \rangle$ such that $\sum_{n \geq 1} \mathsf{length}(J_n) < \varepsilon$ and $E \subseteq \bigcup_{n \geq 1} J_n$. Show that the Cantor set has measure zero.

(35) Let $\langle a_n : n \geq 1 \rangle$ be a bounded sequence of reals. Show that $\langle a_n : n \geq 1 \rangle$ is convergent iff

$$\limsup_{n \to \infty} a_n = \liminf_{n \to \infty} a_n$$

(36) Assume $\limsup_{n \to \infty} x_n < \beta$. Show that there exists $N \geq 1$ such that for all $n \geq N$, $x_n < \beta$.

(37) Assume $\limsup_{n \to \infty} x_n > \beta$. Show that there exist infinitely many n such that $x_n > \beta$.

(38) Show that if $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n \to \infty} a_n = 0$.

(39) Construct a sequence $a_n > 0$ such that $\limsup_{n \to \infty} \frac{a_{n+1}}{a_n} = \infty$ and $\sum_{n=1}^{\infty} a_n$ converges.

(40) Construct a sequence $a_n > 0$ such that $\lim_{n \to \infty} \frac{a_{n+1}}{a_n} = \lim_{n \to \infty} a_n^{1/n} = 1$ and $\sum_{n=1}^{\infty} a_n$ converges/diverges.

(41) Let $a_n > 0$ for every $n \geq 1$. Show that

$$\limsup_{n \to \infty} a_n^{1/n} \leq \limsup_{n \to \infty} \frac{a_{n+1}}{a_n}$$

(42) Describe $\{x : \sum_{n=1}^{\infty} a_n x^n \text{ converges}\}$ in each one of the following cases.

(a) $a_n = n^n$.

(b) $a_n = 1/n!$.

(c) $a_n = 2^n/n^2$.

(d) $a_n = n^3/3^n$.

(43) Suppose $a_n > 0$ for all $n \geq 1$. Show that $\sum_{n=1}^{\infty} a_n$ converges iff $\sum_{n=1}^{\infty} \frac{a_n}{1+a_n}$ converges.

(44) Suppose $f : \mathbb{R} \to \mathbb{R}$ is differentiable on $(x-h, x+h) \setminus \{x\}$ for some $h > 0$ and continuous at x. Assume that $\lim_{y \to x} f'(y) = L$. Show that f is differentiable at x and $f'(x) = L$.

(45) Suppose $f, g : [a, b] \to \mathbb{R}$ are both continuous on $[a, b]$ and differentiable on (a, b). Show that there exists $x \in (a, b)$ such that

$$(f(b) - f(a))g'(x) = (g(b) - g(a))f'(x).$$

(46) Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $|f(x) - f(y)| \leq |x - y|^2$ for every $x, y \in \mathbb{R}$. Show that f is constant.

(47) Let $f : (a, b) \to \mathbb{R}$ be differentiable on (a, b). Assume that $\sup(\{|f'(x)| : x \in (a, b)\}) < \infty$. Show that f is uniformly continuous on (a, b).

(48) Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} e^{-1/x^2} & \text{if } x \neq 0. \\ 0 & \text{if } x = 0. \end{cases}$$

Show that $f \in C^{\infty}(\mathbb{R})$. Compute the Taylor series of f about $x = 0$.